

Personal Info

+306984392304

angeloskatsampekis@gmail.com

Nationality

Greek

Date of birth

19/11/1994

Links

[Portfolio](#)

[LinkedIn](#)

Skills

Unity 3D / C# (HDRP, URP, XR Interaction)

VR/AR development (Meta Quest, Valve Index, OpenXR, Meta XR SDKs)

Game Development

Multiplayer networking (Netcode for GameObjects)

Multithreaded physics & gameplay systems

ML-Agents & Generative-AI pipelines

ONNX, NNAPI, TensorFlowLite, Sentis

Cesium 3D Tiles & Geospatial data

C++, Python tooling, Git, CI/CD

Mobile Optimization: GLES, Vulkan, Android NDK

Data Visualization

Angelos-Ioannis Katsampekis

Unity developer and research engineer

I'm a **Unity Developer** and **XR Research Engineer** with 5+ years of experience creating games and immersive experiences. I've worked on projects for **Samsung, Coca-Cola, and EU Horizon programs**, ranging from VR avatar-tracking systems and AI-powered museums to a full drone-and-missile simulator with advanced physics and ML-based navigation. My background includes strong experience with **Unity, C#, C++, and Python**, as well as practical work in gameplay programming, physics simulation, multiplayer networking, machine learning, generative AI, and performance optimization.

Work Experience

Research Assistant, CERTH - Center for Research & Technology, Hellas, Thessaloniki

2022 - 2025

Drone & Missile Simulator (Unity)

- **Designed** and built the entire **simulator** end-to-end, from core architecture to UX.
- **Developed** and optimized a custom real-time, multithreaded **aerodynamic solver** with per-vertex force computation for accurate flight simulation.
- **Trained** autonomous drones with **Unity ML-Agents** to navigate dense forest environments using a depth camera.
- **Generated and logged** missile-launch telemetry to build custom datasets for **AI targeting research**.

Samsung Electronics - Avatar Body Capture (Unity)

- **Prototyped** a VR full-body avatar estimation system trained on **motion-capture data (AMASS)**, enabling **real-time animation** in VR from only **head and hand inputs**.
- **Integrated multiple inference engines** (Sentis, ONNX Runtime, TensorFlow Lite) to **optimize performance** across XR devices.

Palimpsisto - 3D Geospatial Platform (Unity)

- **Developed** an interactive **geospatial platform** for the Archaeological site of Kythnos, funded by the Greek government.
- **Integrated Cesium** to stream photogrammetry scans as 3D tiles from a Geospatial Database.
- **Connected** an **AI-Assistant** for in-platform interaction.
- **Linked** live museum findings **metadata** allowing curators to update exhibits.

ReEvaluate - AI-Driven VR Museum (EU Horizon, Unity)

- **Developed** an immersive museum where **generative AI** creates themed textures and auto-places artifacts based on semantic descriptions.

5G VR Automotive Tour (Unity Multiplayer)

- **Developed** a VR Multiplayer app in Unity that receives a 360-degree video stream over 5G, applies foveated rendering and face anonymization on the fly and reports networking metrics.

VirTourArt Platform (EU Horizon, Unity SDK)

- **Delivered** AR/VR rendering & multimodal interaction modules, localization & 3D reconstruction and packaged the subsystems into an **SDK**.

EMBNOESIS - Intelligent NB-IoT sensors for the construction industry

- **Developed** a Python **data-visualization tool** that converts live NB-IoT sensor streams into interactive dashboards for construction-site monitoring.

Coca-Cola - CCH: Software Development Service

- **Designed** and implemented machine learning **algorithms** for Coca-Cola, working alongside their data science team to support inventory optimization research.

Software Developer, Pragma - IoT Solutions, Thessaloniki 2024 - 2025

Samsung Electronics - Game Optimization

- **Created** Python and C++ **tools** for Samsung to convert and package ONNX models to ANN for mobile AI runtime, streamlining deployment on Android devices.
- **Benchmarked** mobile **games** with a custom graphics monitoring harness, capturing frame-time, thermal, and battery data to guide optimization in collaboration with Samsung's engineering team.

Education

Physics, Aristotle University of Thessaloniki (BSc) 2020

Advanced Computer and Communication Systems, Aristotle University of Thessaloniki (MSc) 2025

Data Science with Python, National and Kapodistrian University of Athens (Certificate) 2021